

Differentiation & Linear Algebra

A quick-reference formula sheet

I Differentiation — Core Rules

Definition

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Basic Rules

$$(f \pm g)' = f' \pm g' \quad (\text{sum rule})$$

$$(fg)' = f'g + fg' \quad (\text{product rule})$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2} \quad (\text{quotient rule})$$

$$(f \circ g)'(x) = f'(g(x)) g'(x) \quad (\text{chain rule})$$

$$\frac{d}{dx} [f^{-1}(x)] = \frac{1}{f'(f^{-1}(x))} \quad (\text{inverse function rule})$$

Power, Exponential & Logarithmic

$$\frac{d}{dx} [x^n] = nx^{n-1}$$

$$\frac{d}{dx} [a^x] = a^x \ln a$$

$$\frac{d}{dx} [\log_a x] = \frac{1}{x \ln a}$$

$$\frac{d}{dx} [e^x] = e^x$$

$$\frac{d}{dx} [\ln x] = \frac{1}{x}$$

$$\frac{d}{dx} [x^x] = x^x (\ln x + 1)$$

Trigonometric

$$\frac{d}{dx} [\sin x] = \cos x$$

$$\frac{d}{dx} [\tan x] = \sec^2 x$$

$$\frac{d}{dx} [\sec x] = \sec x \tan x$$

$$\frac{d}{dx} [\cos x] = -\sin x$$

$$\frac{d}{dx} [\cot x] = -\csc^2 x$$

$$\frac{d}{dx} [\csc x] = -\csc x \cot x$$

Inverse Trigonometric

$$\frac{d}{dx} [\arcsin x] = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} [\arctan x] = \frac{1}{1+x^2}$$

$$\frac{d}{dx} [\arccos x] = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} [\text{arccot } x] = \frac{-1}{1+x^2}$$

Hyperbolic

$$\frac{d}{dx}[\sinh x] = \cosh x$$

$$\frac{d}{dx}[\tanh x] = \operatorname{sech}^2 x$$

$$\frac{d}{dx}[\cosh x] = \sinh x$$

II Differentiation — Advanced Topics

Higher-Order & Series

$$\text{Leibniz rule: } (fg)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n-k)}$$

$$\text{Taylor series: } f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

$$\text{Faà di Bruno's formula: } \frac{d^n}{dx^n} f(g(x)) = \sum_{k=1}^n f^{(k)}(g(x)) B_{n,k}(g', g'', \dots, g^{(n-k+1)})$$

Implicit & Parametric Differentiation

Implicit: differentiate both sides of $F(x, y) = 0$ treating $y = y(x)$, then solve for $\frac{dy}{dx}$.

$$\frac{dy}{dx} = -\frac{\partial F / \partial x}{\partial F / \partial y}$$

Parametric ($x = x(t)$, $y = y(t)$):

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad \frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{dx/dt}$$

Partial & Multivariable

$$\text{Gradient: } \nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right)$$

$$\text{Total derivative: } df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i$$

$$\text{Multivariable chain rule: } \frac{\partial f}{\partial t} = \sum_{i=1}^n \frac{\partial f}{\partial x_i} \frac{\partial x_i}{\partial t}$$

$$\text{Jacobian matrix: } J = \left[\frac{\partial f_i}{\partial x_j} \right]_{i,j}, \quad \text{Hessian: } H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$$

$$\text{Directional derivative: } D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}, \quad |\mathbf{u}| = 1$$

Mean Value & Error

$$\text{Mean Value Theorem: } f'(c) = \frac{f(b) - f(a)}{b - a}, \quad c \in (a, b)$$

$$\text{L'Hôpital's Rule: } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \quad (\text{if } \frac{0}{0} \text{ or } \frac{\infty}{\infty})$$

$$\text{Newton's method: } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

III Linear Algebra — Matrices & Determinants

Matrix Operations

$$(AB)^T = B^T A^T, \quad (AB)^{-1} = B^{-1} A^{-1}, \quad (A^{-1})^T = (A^T)^{-1}$$
$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

Determinants

For a 2×2 matrix: $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$

For a 3×3 matrix (Sarrus' rule / cofactor expansion):

$$\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

General cofactor (Laplace) expansion along row i :

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} M_{ij}$$

Key properties:

$$\begin{aligned} \det(AB) &= \det(A) \det(B) & \det(A^T) &= \det(A) \\ \det(A^{-1}) &= \frac{1}{\det(A)} & \det(cA) &= c^n \det(A) \quad (n \times n) \end{aligned}$$

Trace

$$\operatorname{tr}(A) = \sum_{i=1}^n a_{ii}, \quad \operatorname{tr}(AB) = \operatorname{tr}(BA), \quad \operatorname{tr}(A) = \sum_i \lambda_i$$

Eigenvalues & Eigenvectors

$$A\mathbf{v} = \lambda\mathbf{v} \iff \det(A - \lambda I) = 0 \quad (\text{characteristic equation})$$

$$\det(A) = \prod_{i=1}^n \lambda_i, \quad \operatorname{tr}(A) = \sum_{i=1}^n \lambda_i$$

$$\text{Diagonalization: } A = PDP^{-1}, \quad D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$$

Cayley–Hamilton theorem: if $p(\lambda) = \det(A - \lambda I)$, then $p(A) = 0$

IV Linear Algebra — Vector Spaces & Decompositions

Vectors & Inner Products

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta, \quad \|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}}$$

$$\text{Cauchy-Schwarz:} \quad |\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

$$\text{Cross product } (\mathbb{R}^3): \quad \mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

$$\text{Projection:} \quad \text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}$$

Rank & Nullity

$$\text{rank}(A) + \text{nullity}(A) = n \quad (\text{rank-nullity theorem, } A: \mathbb{R}^n \rightarrow \mathbb{R}^m)$$

$$\text{rank}(A) = \text{rank}(A^T) = \dim(\text{col}(A)) = \dim(\text{row}(A))$$

Orthogonality & Gram-Schmidt

Given linearly independent $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, construct orthogonal $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$:

$$\mathbf{e}_k = \mathbf{v}_k - \sum_{j=1}^{k-1} \frac{\mathbf{v}_k \cdot \mathbf{e}_j}{\mathbf{e}_j \cdot \mathbf{e}_j} \mathbf{e}_j$$

Matrix Decompositions

$$\text{LU decomposition:} \quad A = LU$$

$$\text{QR decomposition:} \quad A = QR, \quad Q^T Q = I$$

$$\text{Singular Value Decomposition:} \quad A = U \Sigma V^T$$

$$\text{Spectral theorem (symmetric } A): \quad A = Q \Lambda Q^T, \quad Q^T Q = I$$

$$\text{Jordan normal form:} \quad A = P J P^{-1}, \quad J = \text{diag}(J_1, \dots, J_k)$$

Linear Systems

$$A\mathbf{x} = \mathbf{b} \implies \mathbf{x} = A^{-1}\mathbf{b} \quad (\text{if } A \text{ invertible})$$

$$\text{Cramer's rule:} \quad x_i = \frac{\det(A_i)}{\det(A)}, \quad A_i = A \text{ with column } i \text{ replaced by } \mathbf{b}$$

$$\text{Least squares solution:} \quad \hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$$

Quick-reference sheet for differentiation and linear algebra. Not exhaustive — consult a textbook for full derivations and proofs.